ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

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Abstract

Length extrapolation in language models involves competing objectives: retrieval fidelity, long-document likelihood, short-context quality, and inference cost. We present ATMA, a 378M-parameter hybrid recipe that combines Polar Attention with gated-delta recurrent memory, and study these objectives as a Pareto problem rather than claiming general architectural dominance. Polar Attention separates a normalized direction channel from a bounded participation-ratio magnitude channel. We select the recipe with a complete 120-cell, 1B-token factorial sweep, then train matched NoPE, RoPE, and Polar variants for 9.816B tokens at length 2K and evaluate them through 256K. Across the factorial, memory improves Polar's 64K retrieval score in all 20 matched cells (mean +47.8 points), whereas its effect on NoPE is small and inconsistent. At 256K, Polar retains 34.4% teacher-forced target-token accuracy and 9.0% exact five-token accuracy; exact retrieval is 18.0% on synthetic contexts but 0.0% on FinePDFs contexts. Polar also limits mean fixed-target bits-per-byte degradation to 1.26×, at a 1.9-point mean cost on eight short-context tasks. Raven baselines lead BABILong and have length-independent decode state, illustrating a different point on the frontier. Finally, a post-hoc audit finds isolated near-unit retention gates with zero-input half-lives of 21.0M tokens in NoPE and 3.07M in Polar. Capping one runtime head at a 256-token half-life, without changing checkpoint weights, reduces mean 256K BPB from 8.097 to 1.595 for NoPE and from 1.825 to 1.501 for Polar. NoPE BABILong recovers from 0% to 39% and Polar from 28% to 42%; Polar retrieval rises from 34.4/9.0% token/exact to 47.1/16.3%. RoPE does not recover retrieval. We interpret excessive retention as a major checkpoint-specific mediator, not a complete explanation or a validated training method.

1 Introduction

Large language models increasingly process code repositories, books, legal documents, and scientific papers that exceed their pretraining sequence length. Full softmax attention preserves token-level access in principle, but its probability mass can disperse as the key set grows (Nakanishi, 2025; Velickovic et al., 2025; Barbero et al., 2024). Sliding windows bound cost by removing older tokens from direct access. Recurrent sequence models instead keep fixed-size state but compress history. Long-context modeling therefore exposes a multi-objective trade-off among retrieval fidelity, sequence likelihood, reasoning, short-context quality, and serving cost.

We present ATMA, a hybrid architecture that combines short gated convolutions with four Polar Attention layers as global mixers. Polar Attention represents "what matched" with a normalized direction and "how much matched" with a bounded participation-ratio channel. A learned null sink follows the extreme-value scale of irrelevant scores. Each global layer also contains gated-delta memory, providing a compressed recurrent history alongside full-context attention (Appendix A).

Our contributions are threefold. First, a complete 120-cell, 1B-token factorial sweep measures interactions among attention core, recurrent memory, training-time local windows,

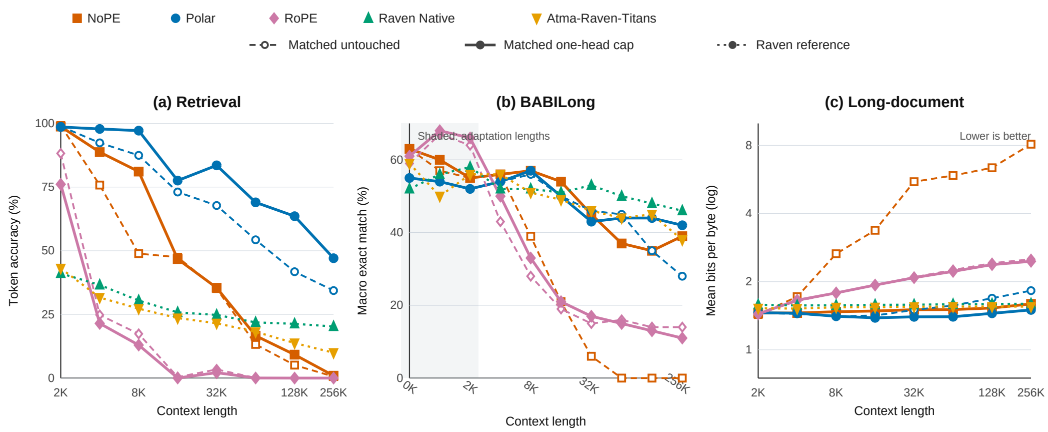

Figure 1: Full length-wise re-evaluation with both recurrent-family comparisons retained. For matched NoPE, Polar, and RoPE, dashed open curves are untouched checkpoints and solid curves apply the inference-only one-head cap. Dotted filled curves are untouched Raven Native and Atma-Raven-Titans references; they are not clamped or optimizer-matched ablations. Left: teacher-forced target-token retrieval. Center: BABILong exact match after adaptation. Right: mean fixed-target BPB (log scale; lower is better).

distractor alignment, and representation regularization. Second, we promote one recipe to matched 9.816B-token NoPE, RoPE, and Polar runs and evaluate retrieval, BABILong, fixed-target likelihood, short-context tasks, and serving. Raven models provide a strong but separately optimized recurrent-family comparison (Afzal et al., 2026); the untouched results locate the original quality–systems frontier. Third, we trace checkpoint-specific extrapolation failure from near-identical 2K loss curves to isolated near-unit retention gates, then test that mechanism with a single-head, inference-only intervention and a full re-evaluation. Figure 1 shows the recovery over every evaluated length rather than only at 256K. The selective recovery identifies excessive retention as a causal mediator in NoPE and Polar, while negative controls and residual failures delimit the claim.

2 Related Work

Softmax attention (Vaswani et al., 2017) has quadratic compute and can lose selectivity as the number of keys grows. RoPE (Su et al., 2021) and ALiBi (Press et al., 2021) alter positional information while retaining dense simplex normalization. Scalable-Softmax, adaptive-temperature analyses, and sparse alternatives address dispersion or length-dependent sharpness (Nakanishi, 2025; Velickovic et al., 2025; Vasylenko et al., 2026). Threshold Differential Attention (TDA) uses a $\sqrt{\log n}$ threshold motivated by maxima of noise scores (Huang et al., 2026). Polar combines length-dependent temperature and extreme-value scaling with a differentiable null sink and bounded magnitude channel. We compare Polar with matched NoPE and RoPE controls; comparisons with these other methods are conceptual rather than empirical.

Linear attention, state-space models, and gated recurrences exchange full-resolution token access for compressed fixed-size history (Gu & Dao, 2023; Yang et al., 2024b;a; Behrouz et al., 2025). Near-unit gates create very long timescales and can enter saturation regimes (Gu et al., 2020); stability-aware SSM parameterizations similarly distinguish stable eigenvalues from robust distance to the stability boundary (Wang & Li, 2024). Stuffed Mamba directly studies an inability-to-forget failure and post-training interventions that restore forgetting (Chen et al., 2025). Our audit asks the corresponding checkpoint-level question for the gated-delta branch used here: a sigmoid keeps each retention value below one, but does not enforce a useful uniform margin from one. Raven combines routed sparse memory, decay dynamics, and gated-delta recurrence (Afzal et al., 2026). We treat it as a strong sub-quadratic baseline, but not as an attention ablation because it uses a different model family and optimizer. Hybrid models such as LFM2 combine local and global mixers for efficiency (Hasani et al., 2025; Allen-Zhu, 2025). ATMA follows this hybrid pattern while changing the global mixer and adding a recurrent memory stream.

3 Method

3.1 Hybrid architecture

ATMA is a 16-layer decoder whose block order repeats local–local–global–local four times, yielding 12 LFM2-style gated convolutional layers and global attention at blocks 3, 7, 11, and 15. All blocks use pre-normalization and squared-ReLU MLPs. The attention variants share hidden size 1024, eight query heads, two key-value heads, head dimension 128, tokenizer, data order, optimizer, schedule, and training budget; only the attention core changes among NoPE, RoPE, and Polar. Canon depthwise causal convolutions are applied to q, k, and v before attention (Allen-Zhu, 2025).

3.2 Polar attention

For query i and its $n_i = i+1$ visible keys, let $\sigma_{ij} = q_i^\top k_j/\sqrt{d_k}$. Each head learns a length temperature and a virtual null logit,

$$t_i = 1 + \mathrm{softplus}(\alpha)\log n_i, \qquad \nu_i = b + \mathrm{softplus}(\gamma)\sqrt{\log(n_i+1)}. \tag{1}$$

The second term tracks the extreme-value scale of unrelated scores. Appending the null as a softmax competitor gives

$$w_i = \mathrm{softmax}(t_i[\sigma_{i\bullet}, \nu_i]), \qquad s_i = \sum_j w_{ij} v_j + w_{iN} v_{\mathrm{null}}. \tag{2}$$

The direction channel is $c_i = s_i/\max(\|s_i\|_2, \varepsilon)$, hence $\|c_i\|_2 \le 1$. To retain bounded information about match multiplicity, real-key weights are renormalized as $\hat{w}_{ij} = w_{ij}/\sum_k w_{ik}$ and summarized by the participation ratio

$$n_{\mathrm{eff},i} = \left(\sum_j \hat{w}_{ij}^2\right)^{-1}, \quad m_{\mathrm{eff},i} = n_{\mathrm{eff},i}(1-w_{iN}), \quad \mu_i = \tanh(\mathrm{softplus}(\beta)\log(1+m_{\mathrm{eff},i})). \tag{3}$$

Thus $\mu_i \in [0,1)$, while the direction channel removes radial scale. This is a channel decomposition, not an independence claim: changing the match set can rotate c_i. The Polar output is a gated projection of c_i plus a learned projection of μ_i. These choices were selected through pre-integration falsification tests: additive soft counts accumulated leakage with length, fixed or relative floors were overtaken by noise tails, and an unbounded $\log m$ channel left the training range. The standalone prototype was not retained; Appendix C distinguishes this design provenance from the reproducible integrated oracle and kernel tests. A distractor-alignment auxiliary loss was tested in the factorial but is disabled in the promoted recipe.

3.3 Gated-delta memory

Each global layer also maintains a per-head recurrent state $M_t \in \mathbb{R}^{d_v \times d_k}$. With unit-normalized keys and queries, retention $\gamma_t = \sigma(W_\gamma x_t + b_\gamma)$, and write gate β_t, the decay-first, self-inclusive recurrence is

$$M_t = \gamma_t M_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top, \qquad r_t = M_t q_t. \tag{4}$$

RMS-normalized r_t is gated, projected, and added to the Polar output. The projection is zero-initialized. Unit L_2 normalization is important: RMS-normalized keys have $\|k\|_2^2 = d_k$ and made the tested delta recurrence diverge, whereas unit keys keep the update eigenvalue along k within the gate-controlled range. Under bounded values and a uniformly contractive

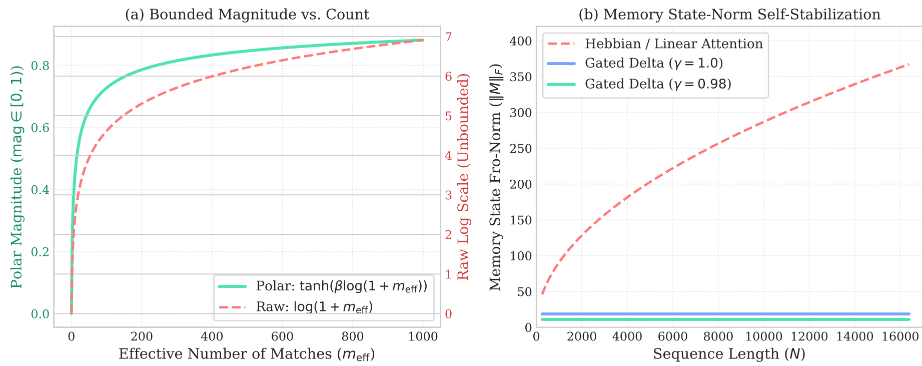

Figure 2: ATMA diagnostics. Left: the effective match count is mapped to a bounded magnitude channel. Right: the tested gated-delta state remains flat with length, unlike an ungated Hebbian memory. These controlled diagnostics illustrate the mechanism, they are not a network-level stability proof.

retention gate, Appendix A gives a sequence-length-independent state bound. The sigmoid guarantees $0 < \gamma_t < 1$ pointwise, but not the fixed margin $\gamma_t \leq \bar{\gamma} < 1$ assumed by that result: the implied half-life $H_{1/2} = \log(1/2)/\log\gamma$ diverges as $\gamma \to 1$. Section 6 audits this distinction empirically.

Figure 2 summarizes the two mechanisms and their controlled diagnostics. Appendix B defines every regularization mode in the factorial sweep, Appendix C gives the numerical oracle and streaming Triton implementation, and Appendix H details the kernel and serving optimizations.

4 Experimental Design

4.1 Recipe selection

Stage I trains all $3 \times 5 \times 2 \times 2 \times 2 = 120$ combinations of attention type (Polar, RoPE, NoPE), regularization, distractor loss, memory, and a 1024-token training window. Every model receives approximately 1B FineWeb-Edu tokens at length 2048 on an NVIDIA L4, and every evaluation uses full context. These are distinct factorial cells rather than seed replicates; Stage I selects the promoted recipe and is not used for headline absolute comparisons.

4.2 Matched comparison

Stage II trains 378.16–378.22M-parameter NoPE, RoPE, and Polar models for 9.816B tokens at length 2048 in the same L40S software and hardware environment. All three use the same Muon/AdamW parameter split. Raven Native (388.54M) and Atma-Raven-Titans (382.37M) use the same data, tokenizer, token budget, and device class, but use AdamW. This follows a 1B-token Raven Native pilot with the ATMA Muon schedule: validation loss improved to 3.76 nats at step 625, then destabilized and ended at 4.62, with 0% retrieval through 64K. We therefore report the stable Raven models as a separate model-family/optimizer group. The single failed pilot does not establish incompatibility with every Muon configuration.

Evaluation. The evaluation contains 24,000 paired retrieval trials over synthetic and FinePDFs haystacks, lengths 2K–256K, depths 10/50/90%, and 50 trials per cell; BABILong adaptation and exact match (Kuratov et al., 2024); fixed-target bits per byte (BPB) on FinePDFs, PG-19, and Proof-Pile; eight likelihood-scored downstream tasks (22,939 examples per model); and single-sequence prefill/decode on one L40S. Retrieval teacher-forces the five-token target and reports both token accuracy and exact five-token accuracy. Neither metric is autoregressive generation success. BABILong uses custom splits and prompts, so

64K matched contrast	Mean Δ	Median Δ	Range	Positive
Polar: memory on − off	+47.8	+51.3	$[+7.5, +92.5]$	20/20
NoPE: memory on − off	+4.6	+1.9	$[-15.0, +21.3]$	11/20
RoPE: memory on − off	0.0	0.0	$[0.0, 0.0]$	0/20
Polar+memory − NoPE+memory	+41.7	+41.9	$[0.0, +77.5]$	19/20 (+1 tie)
Polar+memory − RoPE+memory	+50.3	+51.9	$[+18.8, +92.5]$	20/20

Table 1: Matched contrasts across the 120-cell, 1B-token factorial. Values are percentage-point changes in teacher-forced five-token accuracy at 64K. Each row holds all other factorial axes fixed. "Positive" excludes one numerical tie in the Polar–NoPE comparison.

Group	Model	Retrieval (%)			BABILong (%)		BPB		
		Tok. 2K	Tok. 256K	Exact 256K	2K	256K	2K	256K	ratio
Matched	NoPE	99.2	0.6	0.0	55	0	1.432	8.097	5.65×
Matched	Polar	98.9	34.4	9.0	52	28	1.451	1.825	1.26×
Matched	RoPE	88.2	0.0	0.0	64	14	1.439	2.512	1.75×
Raven/AdamW	Atma-Raven-Titans	43.1	10.0	0.0	56	38	1.525	1.551	1.02×
Raven/AdamW	Raven Native	41.1	20.3	0.0	58	46	1.581	1.597	1.01×

Table 2: Long-context endpoints. Retrieval averages tasks, suites, and depths; "Tok." is teacher-forced target-token accuracy and "Exact" requires all five target tokens to be correct. BABILong is macro exact match. BPB averages three fixed-target datasets; lower is better. Bold is best within each comparison group.

its absolute scores are internally comparable rather than leaderboard replications. Protocols, denominators, and complete matrices appear in Appendix D.

5 Results

5.1 Component selection

Table 1 summarizes matched contrasts from the full factorial rather than selecting individual cells. At 64K, adding memory improves Polar in all 20 combinations of regularizer, distractor setting, and window setting, by 47.8 points on average. The corresponding NoPE effect is small and heterogeneous, and RoPE remains at zero. Among memory-enabled cells, Polar is never worse than NoPE and exceeds RoPE in all 20 comparisons. A training window reduces Polar+memory by 25.5 points on average and is harmful in 8/10 matched cells; distractor alignment has a smaller, nonuniform effect (mean −6.0 points; harmful in 6/10). We therefore promote full-context Polar plus memory without distractor alignment. The selected baseline cell reaches 92.5% token accuracy at 64K; the complete matrix is in Appendix E.

5.2 A multi-objective frontier

Table 2 reports extrapolation endpoints. At 2K, NoPE and Polar achieve nearly perfect target-token accuracy. At 256K, NoPE and RoPE fall to 0.6% and 0.0%, while Polar retains 34.4% averaged across tasks, suites, and depths. Exact five-token accuracy gives a stricter boundary: Polar reaches 9.0% overall at 256K, entirely from synthetic haystacks (18.0% synthetic; 0.0% FinePDFs). At 64K, Polar obtains 65.7% exact accuracy on synthetic contexts but 0.0% on FinePDFs, despite 16.3% target-token accuracy there. We therefore claim retained target signal and exact synthetic retrieval, not successful real-text retrieval at those lengths. Full token, exact, depth, and haystack matrices are in Appendix F.

Table 3 separates target-token accuracy by haystack and length; Appendix F reports the corresponding exact-sequence values. Polar also limits mean BPB degradation to 1.26× at 256K, compared with 5.65× for NoPE and 1.75× for RoPE. This stability accompanies a short-context cost: mean accuracy over eight 2K controls is 43.29% for Polar, 45.15%

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 3: Zero-shot retrieval accuracy (%) by haystack type. Values average teacher-forced token accuracy over needle depths. The recurrent hybrid is omitted for space and reported in Appendix F.

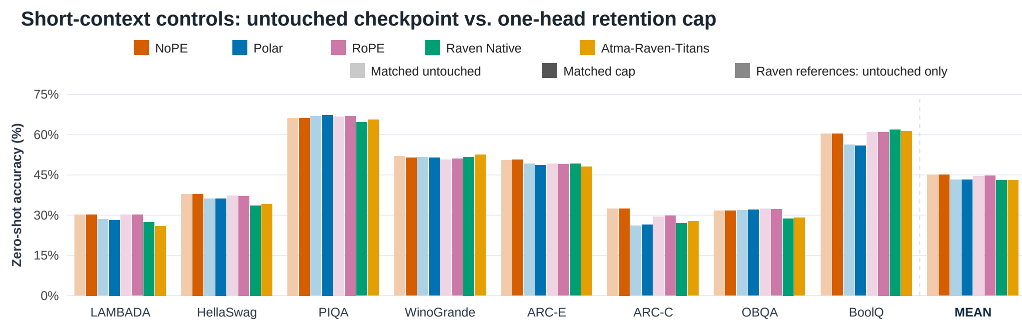

Figure 3: Zero-shot accuracy on eight 2K language-modeling controls (22,939 examples per model). For the matched attention variants, pale bars are untouched and saturated bars are capped. Raven Native and Atma-Raven-Titans appear as saturated untouched references only. The rightmost group gives unweighted task means; exact paired matched-model values are in Table 19.

for NoPE, and 44.60% for RoPE. Figure 3 shows that the gap is task-dependent rather than uniform. The 1.86-point mean gap to NoPE prevents a blanket quality claim. Exact per-task values are in Appendix G.

Table 4 makes the quality–systems trade-off explicit. Raven Native reaches 46% BABILong at 256K and keeps BPB nearly flat, while its fixed recurrent state avoids a sequence-length-dependent KV cache. At 128K, Atma-Raven-Titans decodes at 2.27 ms/token versus 16.3 ms/token for Polar; Polar instead has higher synthetic retrieval and training MFU. The BABILong samples are small: simple Wilson intervals at 256K are 20.1–37.5% for Polar (28/100) and 36.6–55.7% for Raven Native (46/100). Because Raven also changes model family and optimizer, these rows locate practical operating points rather than isolate one causal mechanism.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 4: Short-context and systems trade-offs. Serving uses one sequence and one timing sample on one L40S, so the values are descriptive rather than uncertainty estimates.

6 Checkpoint Variability and Retention-Horizon Audit

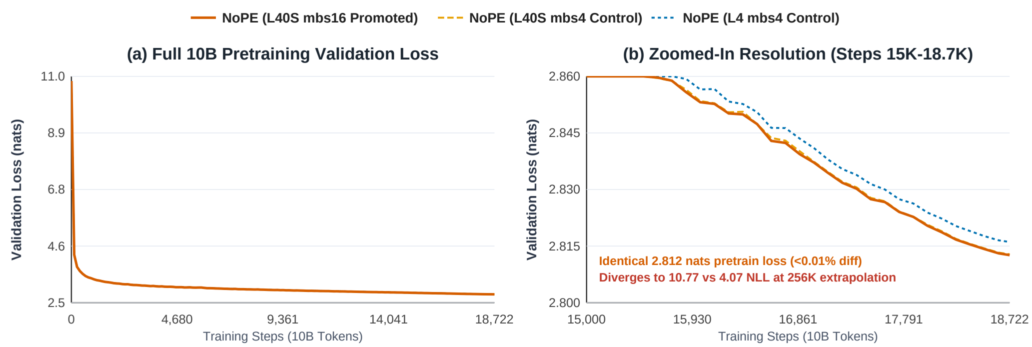

Figure 4: NoPE 2K validation trajectories are nearly coincident, yet the checkpoints' 256K NLL values are 10.77, 9.90, and 4.07. The two L40S runs differ in microbatch size; the L4 run differs in device class.

Figure 4 shows the trigger for a post-hoc audit. Three NoPE runs finish at 2.8126, 2.8128, and 2.8160 validation nats at 2K, yet reach 10.77, 9.90, and 4.07 NLL at 256K. The runs were not initialized from a recorded common seed, so the comparison establishes checkpoint variability, not a device-class treatment effect. Evaluation replays, a microbatch control, and deterministic fixtures ruled out several immediate explanations but did not identify the origin (Appendix K).

6.1 Parameter inspection

We next inspected the retention branch rather than sampling whole-block gains. For each layer and query/memory head, we evaluate the learned zero-input operating point $\gamma_0 = \sigma(b_\gamma)$ and convert it to $H_{1/2} = \log(1/2)/\log\gamma_0$. Although the attention path has two KV heads, the memory gate has eight independently learned query/memory-head rows. The promoted NoPE checkpoint contains one block-2 head with a 21.01M-token half-life; Polar contains one at 3.07M tokens. The corresponding L4 NoPE maximum is 185 tokens, while Atma-Raven-Titans stays between 30.5 and 41.4 tokens. Nonzero W_γ rows mean runtime retention remains activation-dependent, so this scan is a diagnostic operating point rather than a complete trace. The same outlier persists after BABILong fine-tuning (20.86M NoPE; 3.05M Polar).

6.2 Single-head intervention and full re-evaluation

To test causality, we cap only the final runtime logit of the largest parameter-only head at

$$\gamma_{\max} = 2^{-1/256} = 0.997296, \tag{5}$$

which imposes a 256-token maximum half-life without rewriting checkpoint tensors or changing attention. In a same-process paired sweep, the baseline and cap share the model instance, documents, needle trials, and attention backend. At 256K, clean NLL changes from 12.003 to 1.138 for the promoted NoPE checkpoint and from 1.252 to 1.057 for Polar; needle cross-entropy changes from 12.20 to 6.05 and from 2.39 to 1.71, respectively. RoPE changes only from 6.16 to 6.02 needle cross-entropy and loses short-range retrieval, providing a negative control. Appendix I reports every checkpoint and endpoint.

We then reran downstream, retrieval, fixed-target BPB, and BABILong for the three matched attention variants. Figure 1 plots the complete length curves, while Table 5 gives the endpoints and Appendix I gives every numerical cell. These clamped runs are compared with the pinned archived baseline matrices; the paired pilot above, rather than this cross-run comparison, supplies the controlled intervention. All 15 requested jobs completed without an out-of-memory failure.

Model	Max $H_{1/2}$	Base mean	256K BPB	256K retrieval tok./exact	256K BABI
NoPE	21.01M	45.15 → 45.10	8.097 → 1.595	0.60/0.00 → 0.93/0.00	0 → 39
Polar	3.07M	43.29 → 43.25	1.825 → 1.501	34.37/9.00 → 47.07/16.33	28 → 42
RoPE	682	44.60 → 44.66	2.512 → 2.460	0.00/0.00 → 0.00/0.00	14 → 11

Table 5: Inference-only retention diagnostic: archived baseline → one-head 256-token half-life cap. Base mean averages eight 2K tasks (%); retrieval is token/exact accuracy (%); BABI is macro exact match (%). BPB averages three fixed-target datasets. The cap is a probe, not a proposed trained model.

The intervention largely removes NoPE's likelihood collapse and raises BABILong from 21/6/0/0/0% to 54/45/37/35/39% at 16/32/64/128/256K, but exact retrieval remains zero at 256K. This is broad extrapolation recovery, not a single favorable endpoint. For Polar, it improves 256K token retrieval by 12.70 points, exact retrieval by 7.33 points, and BABILong by 14 points; its mean BPB degradation falls from 1.26× to 1.03×. Ordinary 2K downstream means are unchanged within 0.06 points. RoPE's failure to recover retrieval, and its BABILong decrease at 256K, show that the cap is not a generic benchmark booster. Together these results identify excessive retention as a major checkpoint-specific mediator of NoPE and Polar degradation, not the sole cause of extrapolation failure and not evidence that 256 is a universally optimal horizon.

6.3 What this audit cannot answer

The audit leaves several questions unresolved:

1. What training event produces the isolated outlier, and does it replicate under seed-paired device, kernel, and optimizer interventions?
2. Would a bounded train-time parameterization prevent the outlier without removing useful long timescales? The inference cap does not answer this.
3. Why does NoPE exact retrieval remain zero after likelihood and BABILong recover, and why does RoPE not benefit from the same cap?
4. Does Polar's remaining natural-text retrieval gap improve with longer or mixed-domain training, or reflect a synthetic-to-natural induction bias?

These questions delimit the current evidence rather than extend its claims. Resolving them requires interventions that separately vary architecture, optimizer, data, and execution environment.

7 Limitations and Conclusion

This study uses one model scale, one 2K pretraining regime, and no seed replication sufficient for population-level architectural uncertainty. The Stage I factorial demonstrates robustness across nuisance-factor settings, not across random seeds. Teacher-forced retrieval, custom BABILong adaptation, and single-sample serving measurements answer different questions and should not be collapsed into one score. Polar has zero exact FinePDFs retrieval at 64K and 256K in the untouched checkpoint despite retaining partial target-token signal, and Raven is not an optimizer-matched ablation. The recurrent bound requires a uniform learned margin that the sigmoid parameterization does not enforce. The one-head cap diagnoses existing checkpoints; it neither identifies the outlier's training origin nor validates a bounded train-time formulation. Its broad re-evaluation uses pinned archived baselines, with controlled pairing supplied by the smaller sweep. The documented pre-integration Polar design tests are not a substitute for network-level component ablations, and the present comparisons do not cover every contemporary length-scaling method.

Within those limits, ATMA establishes a reproducible operating point for bounded full-context attention plus recurrent memory. Polar materially extends target-token signal, preserves exact synthetic retrieval, and stabilizes long-document likelihood while remaining competitive on short-context tasks. Raven illustrates the BABILong and serving advantages of compressed recurrent state. The checkpoint audit adds a mechanistic qualification: nominally stable gates can sit arbitrarily close to one, and a single extreme retention horizon can dominate extrapolation without appearing in 2K loss. Selective inference-time recovery makes that failure observable and falsifiable while leaving its training origin open. We release configurations, logs, checkpoint manifests, benchmark matrices, clamp specifications, and numerical proof artifacts so that each comparison can be audited.

AI Use Statement

Generative AI tools were used in this revision to assist diagnostic-script implementation, edit and structure the manuscript, and aggregate archived and newly produced evaluation logs. The benchmark jobs themselves were executed by the authors on the listed checkpoints and hardware; AI was not used to generate evaluation datasets or train the reported models. The authors manually compare all AI-assisted mathematical text with the original derivations and proof artifacts, verify empirical values against source logs, and take responsibility for every claim and artifact in the submission. This statement describes the present version as required by the ICLR 2027 policy.

Reproducibility Statement

Model definitions and training settings are described in Sections 3–4. Appendix D specifies evaluation protocols and denominators, Appendix I records the parameter scan, paired intervention, and full re-evaluation, and Appendices E–J provide detailed tables and diagnostics. The anonymous artifact at https://anonymous.4open.science/r/atma contains configurations, logs, checkpoint manifests, benchmark payloads, clamp specifications, and machine-checked proof skeletons.

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A Theoretical Framing

These are sufficient-condition results. They establish noise-odds and norm bounds under explicit assumptions. They do not establish distributional invariance or replace the empirical evidence.

A.1 Extreme-value null floor

Assumption 1 (Sub-Gaussian noise scores). For irrelevant keys $j \in \mathcal{N}_n$, $|\mathcal{N}_n| \le n$, scores X_j are mean-zero and v^2-sub-Gaussian: $\mathbb{E}\exp(tX_j) \le \exp(v^2t^2/2)$ for all $t \in \mathbb{R}$.

Let the Polar null floor be $\nu_n = b + s\sqrt{\log(n+1)}$ with $b, s \ge 0$.

Proposition 1 (Noise exceedances). Under Assumption 1,

$$\mathbb{E}\left[\sum_{j\in\mathcal{N}_n} \mathbf{1}\{X_j > \nu_n\}\right] \le n\exp\left(-\frac{\nu_n^2}{2v^2}\right). \tag{6}$$

If $s \ge \sqrt{2}v$ this is $O(1)$; if the inequality is strict it decays polynomially.

Proof. Apply the sub-Gaussian tail bound to each key and sum. Since $b \ge 0$, $\nu_n^2 \ge s^2\log(n+1)$. □

Unlike TDA, Polar retains sub-threshold keys. With temperature $\theta_n = 1 + a\log n$, define the unnormalized noise-to-null odds $R_n = \sum_{j\in\mathcal{N}_n} \exp(\theta_n(X_j - \nu_n))$.

Proposition 2 (Vanishing null odds). Suppose $s > c > \sqrt{2}v$ and the null floor is $s\sqrt{\log(n+1)}$ up to a nonnegative offset. With probability tending to one,

$$R_n \le n\exp\Big(-(1 + a\log n)(s-c)\sqrt{\log n}\Big). \tag{7}$$

For $a > 0$, the bound converges to zero faster than any inverse polynomial.

Proof. A union bound gives $\Pr[\max_j X_j > c\sqrt{\log(n+1)}] \le n(n+1)^{-c^2/(2v^2)} \to 0$. On the complementary event, every noise score is at least $(s-c)\sqrt{\log n}$ below the floor. Bound each summand and sum over at most n keys. □

A.2 Bounded channels and recurrent state

Proposition 3 (Bounded Polar readout). For $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, $\|c_i\|_2 \le 1$. For $\mu_i = \tanh(\beta \log(1 + m_{\mathrm{eff},i}))$, $\mu_i \in [0, 1)$ when $\beta, m_{\mathrm{eff},i} \ge 0$. A distribution uniform over m real keys has participation ratio m.

Proof. The first two claims follow from the denominator and range of tanh. Uniform weights satisfy $(\sum_{j=1}^{m}(1/m)^2)^{-1} = m$. □

Assumption 2 (Contractive memory). For every t, $\|q_t\|_2 = \|k_t\|_2 = 1$, $\|v_t\|_2 \le V$, $0 \le \beta_t \le 1$, and $0 \le \gamma_t \le \bar{\gamma} < 1$.

Proposition 4 (Length-bounded state). For recurrence equation 4,

$$\|M_t\|_F \le \bar{\gamma}^t \|M_0\|_F + \frac{V}{1 - \bar{\gamma}}. \tag{8}$$

Proof. $I - \beta_t k_t k_t^\top$ has spectral norm at most one and $\|\beta_t v_t k_t^\top\|_F \le V$. Thus $\|M_t\|_F \le \bar{\gamma}\|M_{t-1}\|_F + V$; unrolling gives the result. □

The Lean artifact checks the finite-sum, exponential-monotonicity, bounded-channel composition, and invariant-set induction steps. It takes the sub-Gaussian tail, high-probability margin, and one-step contraction as premises. It is therefore a machine-checked proof skeleton, not end-to-end formal verification. The learned Polar conditions $b \ge 0$ and $s > c > \sqrt{2}v$ require an empirical head-wise audit before they can be asserted for a trained checkpoint. Likewise, $\gamma_t = \sigma(W_\gamma x_t + b_\gamma)$ establishes only $0 < \gamma_t < 1$ at any finite logit; it does not supply one sequence-uniform $\bar{\gamma} < 1$. Appendix I shows that the distinction is practically material: individual zero-input operating points can imply million-token half-lives, although that parameter-only scan does not by itself bound activation-conditioned runtime gates.

B Regularization Modes

The factorial grid varies a representation regularizer applied to the model's signature stream. For every non-baseline mode, the sweep uses the same weight $\alpha_{\mathrm{sig}} = 0.01$ and optimizes

$$\mathcal{L} = (1 - \alpha_{\mathrm{sig}})\mathcal{L}_{\mathrm{LM}} + \alpha_{\mathrm{sig}}\mathcal{L}_{\mathrm{reg}} + \lambda_{\mathrm{dist}}\mathcal{L}_{\mathrm{dist}}. \tag{9}$$

The weak and strong modes follow the two SIGReg families introduced by Weak-SIGReg (Akbar, 2026) and LeJEPA's strong SIGReg objective (Balestriero & LeCun, 2025). The discrete and zipfian modes are local variants that combine those ideas with stronger geometric priors. Table 6 defines all five sweep settings.

Mode	Meaning in the sweep
baseline	No signature-stream regularization; equivalently, $\alpha_{\text{sig}} = 0$.
weak	Weak SIGReg: center a random sketch of the token representations and match its covariance to the identity, $\|\text{Cov}(Sx) - I\|_F$. This encourages decorrelated, unit-variance features at low cost.
strong	Strong SIGReg: project representations onto random one-dimensional directions and match their empirical characteristic functions to the standard Gaussian characteristic function over a fixed frequency grid.
discrete	Local variant: normalize each representation to the sphere of radius $\sqrt{d}$, then apply weak covariance matching. This preserves whitening pressure while encouraging fixed-norm, discrete-code-like directions.
zipfian	Local variant: normalize directions for an orthogonality penalty, then match sorted representation magnitudes to a Zipf profile r^{-1}. This biases the signature stream toward heavy-tailed feature use rather than uniform energy allocation.

Table 6: Regularization modes in the 120-cell factorial sweep. Weak and strong are SIGReg-style covariance and distribution-matching objectives; discrete and zipfian are exploratory local variants.

C Polar Attention Implementation

C.1 Design provenance and scope

Before integration, the Polar choices were developed by falsifying simpler alternatives in a standalone synthetic sandbox. Additive soft counts accumulated small per-key leakage with sequence length; thresholds based on a fixed number of standard deviations retained a constant fraction of noise keys; a fixed null logit was eventually overtaken by the maximum noise score; and an unbounded $\log m$ magnitude exposed the output projection to values beyond its training range. These failures motivated participation ratio, the $\sqrt{\log n}$ null floor, and the bounded $\tanh(\beta \log(1+m))$ map. The standalone prototype was subsequently removed and is not part of the released artifact. We therefore use these tests as design provenance, not as model-scale component ablations or independently reproducible benchmark evidence. The integrated equations, materialized oracle, streaming implementation, and parity tests described below are retained.

C.2 Numerical oracle

The materialized PyTorch path forms the masked real-key logits and virtual null logit in FP32, applies a single softmax, and computes the direction and magnitude channels from the resulting weights. It is used as a numerical oracle for forward and backward tests, not as the production long-context path, because materializing the $T \times T$ score matrix is memory-prohibitive at the evaluated lengths.

C.3 Streaming sufficient statistics

The Triton kernel follows the online-max strategy used by FlashAttention-style kernels. For one query row, it streams K, V blocks and retains only four FP32 statistics: the running maximum M, real-key exponential sum L, squared exponential sum Q_2, and unnormalized value sum S. If a new block changes the row maximum, $\alpha = \exp(M_{\text{old}} - M_{\text{new}})$ rescales each statistic as shown in Table 7.

After the real keys are consumed, the virtual null is folded in as one additional softmax entry. Let $p_0 = \exp(t_i \nu_i - M)$ and $Z = L + p_0$. The kernel obtains

$$c_i = \text{normalize}(S + p_0 v_{\text{null}}), \qquad n_{\text{eff},i} = \frac{L^2}{Q_2}, \qquad m_{\text{eff},i} = n_{\text{eff},i} \frac{L}{Z}. \tag{10}$$

The common factor $1/Z$ cancels under radial normalization of the direction numerator. The production kernel evaluates the magnitude with $2\sigma(2x) - 1 = \tanh x$ and saves (M, L, Q_2, S)

State	Online update	Use after streaming
M	$M \leftarrow \max(M, \max a)$	Stable common logit origin
L	$L \leftarrow \alpha L + \sum_j p_j$	Real-key softmax mass
Q_2	$Q_2 \leftarrow \alpha^2 Q_2 + \sum_j p_j^2$	$n_{\text{eff}} = L^2/Q_2$
S	$S \leftarrow \alpha S + \sum_j p_j v_j$	Direction numerator

Table 7: Streaming state for one Polar query row, where a is the current logit block and $p_j = \exp(a_j - M_{\text{new}})$. The squared accumulator must rescale by α^2.

for the backward pass. Listing 1 gives the algorithmic core; pointer arithmetic, launch metadata, and backward code are omitted.

Listing 1: Schematic core of the block-streamed Triton forward pass.

```
@triton.jit
def polar_fwd_core(q, K, V, n_i, temp, nullv, beta,
                   BLOCK_N: tl.constexpr, DK: tl.constexpr):
    M  = tl.full([BLOCK_M], -1e38, tl.float32)
    L  = tl.zeros([BLOCK_M], tl.float32)
    Q2 = tl.zeros([BLOCK_M], tl.float32)
    S  = tl.zeros([BLOCK_M, DK], tl.float32)

    for start in range(0, n_i, BLOCK_N):
        k, v, valid = load_kv_block(K, V, start, n_i)
        a = tl.where(valid, temp[:, None] * dot(q, k), -1e38)
        M_new = tl.maximum(M, tl.max(a, axis=1))
        alpha = tl.exp(M - M_new)
        p = tl.where(valid, tl.exp(a - M_new[:, None]), 0.0)
        L  = alpha * L + tl.sum(p, axis=1)
        Q2 = alpha * alpha * Q2 + tl.sum(p * p, axis=1)
        S  = alpha[:, None] * S + tl.dot(p, v)
        M = M_new

    # Fold in the learned virtual null as one extra entry.
    M_new = tl.maximum(M, temp * nullv)
    alpha = tl.exp(M - M_new)
    L, Q2, S = alpha * L, alpha * alpha * Q2, alpha[:, None] * S
    p_null = tl.exp(temp * nullv - M_new)
    Z = L + p_null

    direction = normalize(S + p_null[:, None] * v_null)
    n_eff = L * L / tl.maximum(Q2, eps)
    m_eff = n_eff * L / tl.maximum(Z, eps)
    magnitude = 2 * tl.sigmoid(2 * beta * tl.log(1 + m_eff)) - 1
    save_for_backward(M_new, L, Q2, S)
```

This removes the quadratic score-matrix storage. Full-context prefill still performs quadratic query-key arithmetic. The kernel is cross-checked against the materialized PyTorch oracle in FP32 accumulation before lower-precision outputs are accepted.

D Evaluation Protocols

Stage I contains 120 approximately 370–378M-parameter runs trained for 1,900 steps of 524,288 tokens at sequence length 2,048 on FineWeb-Edu. Stage II contains three matched attention models trained for 18,722 such steps (9.816B tokens). The five promoted checkpoints and ten open-weight controls are evaluated with checkpoint-exact forward paths. Stage I cells vary factorial settings and are not independent seed replicates.

D.1 Raven optimizer control

The Raven rows use AdamW optimizer because the attempted matched-schedule control did not produce a usable baseline. In the 1B-token Raven Native pilot with the Atma Muon schedule, validation loss decreased from 10.83 to 3.76 nats by step 625, then rose to 6.88 at step 1,125 and ended at 4.62; teacher-forced retrieval was 0% at every tested length from 2K through 64K. The reported Raven Native and Atma-Raven-Titans checkpoints therefore use AdamW and remain a separate model-family/optimizer group. This is evidence about the tested Muon schedule, not a general claim that Raven cannot converge under any Muon tuning.

Retrieval crosses passkey and NIAH tasks, synthetic and FinePDFs haystacks, eight lengths from 2K through 256K, three depths, and 50 paired trials per cell. The scorer appends the complete five-token target and evaluates each position conditioned on the preceding ground-truth target tokens. Token accuracy is the fraction of correct per-position argmax predictions; exact accuracy requires all five positions to be correct. Both are teacher-forced diagnostics, not autoregressive generation. An older free-generation protocol was retired because all base checkpoints scored uniformly zero. BABILong uses answer-only fine-tuning on tasks qa1–qa10 at 0K/1K/2K and greedy exact match on ten held-out rows per task and length. Base-task controls use LAMBADA, HellaSwag, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA, and BoolQ. Fixed-target BPB uses FinePDFs, PG-19, and Proof-Pile. All reported processes completed without failed, OOM, unsupported, missing, or null cells.

For proportions we report simple Wilson intervals where they aid interpretation. These intervals treat trials as binomial observations and do not exploit the paired design or account for correlations induced by shared templates. The archived retrieval payloads retain aggregate cell counts but the BABILong payloads do not retain per-model correctness vectors, so paired bootstrap or McNemar tests cannot be reconstructed without rerunning or recovering the original predictions.

E Complete Factorial Sweep

Table 8 retains all 120 factorial cells; the main paper reports only the recipe transitions needed to justify promotion.

Table 8: Complete Results of the 120-Run Ablation Sweep. All models are 370M parameters, trained on FineWeb-Edu for 1B tokens ($seq\_len = 2048$). Evaluated from 2K to 64K context length. Clean NLL is measured on coherent FinePDFs documents; junk NLL is measured on the concatenated validation stream. NLL is in nats/token (lower is better), Needle is greedy per-digit accuracy in % (higher is better), and MFU is training model flops utilization in %.

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	baseline	Off	Off	Off	3.323	2.83	3.60	3.27	5.68	96.3	0.0	40.1
POLAR	baseline	Off	Off	On	3.343	2.83	2.66	3.30	4.74	67.5	0.0	39.6
POLAR	baseline	Off	On	Off	3.169	2.70	1.96	3.11	3.13	91.3	92.5	35.6
POLAR	baseline	Off	On	On	3.174	2.71	2.01	3.12	3.13	51.3	30.0	35.6
POLAR	baseline	On	Off	Off	3.332	2.81	5.56	3.28	6.54	85.0	0.0	33.3
POLAR	baseline	On	Off	On	3.361	2.85	2.89	3.35	4.97	91.3	2.5	34.5
POLAR	baseline	On	On	Off	3.178	2.72	1.98	3.12	3.14	73.8	58.8	31.8
POLAR	baseline	On	On	On	3.182	2.73	1.97	3.12	3.14	56.3	21.3	31.2
POLAR	weak	Off	Off	Off	3.333	2.82	2.86	3.28	5.35	88.8	0.0	39.0
POLAR	weak	Off	Off	On	3.348	2.90	3.76	3.38	5.24	93.8	0.0	37.9
POLAR	weak	Off	On	Off	3.179	2.74	1.95	3.12	3.14	70.0	51.3	36.2
POLAR	weak	Off	On	On	3.187	2.74	1.99	3.13	3.15	41.3	41.3	36.2
POLAR	weak	On	Off	Off	3.335	2.82	3.47	3.28	5.56	91.3	0.0	32.8
POLAR	weak	On	Off	On	3.353	2.82	2.58	3.32	4.81	96.3	1.3	32.8
POLAR	weak	On	On	Off	3.178	2.74	1.95	3.12	3.14	33.8	32.5	30.7
POLAR	weak	On	On	On	3.183	2.74	1.94	3.13	3.15	81.3	78.8	30.7
POLAR	strong	Off	Off	Off	3.338	2.85	3.79	3.28	5.83	98.8	1.3	37.3
POLAR	strong	Off	Off	On	3.360	2.85	2.99	3.32	5.01	91.3	0.0	37.9
POLAR	strong	Off	On	Off	3.172	2.72	1.96	3.11	3.13	80.0	52.5	35.8

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Table 8: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	strong	Off	On	On	3.181	2.72	1.94	3.12	3.15	76.3	68.8	34.6
POLAR	strong	On	Off	Off	3.337	2.87	3.22	3.28	5.52	83.8	1.3	32.8
POLAR	strong	On	Off	On	3.347	2.82	2.64	3.30	4.88	97.5	16.3	32.2
POLAR	strong	On	On	Off	3.179	2.73	1.94	3.12	3.14	86.3	83.8	30.9
POLAR	strong	On	On	On	3.182	2.72	1.94	3.12	3.14	25.0	23.8	30.9
POLAR	discrete	Off	Off	Off	3.338	2.82	2.80	3.29	5.12	93.8	2.5	37.9
POLAR	discrete	Off	Off	On	3.349	2.87	2.90	3.33	5.25	93.8	6.3	37.9
POLAR	discrete	Off	On	Off	3.184	2.75	2.03	3.12	3.14	76.3	73.8	36.4
POLAR	discrete	Off	On	On	3.178	2.71	1.92	3.12	3.15	31.3	18.8	35.8
POLAR	discrete	On	Off	Off	3.333	2.80	4.13	3.28	6.12	97.5	0.0	32.6
POLAR	discrete	On	Off	On	3.351	2.87	2.72	3.33	4.89	85.0	2.5	33.3
POLAR	discrete	On	On	Off	3.189	2.74	1.96	3.13	3.15	87.5	53.8	31.8
POLAR	discrete	On	On	On	3.192	2.74	2.07	3.13	3.15	21.3	26.3	31.7
POLAR	zipfian	Off	Off	Off	3.338	2.85	4.83	3.28	6.55	96.3	11.3	37.9
POLAR	zipfian	Off	Off	On	3.350	2.83	2.58	3.32	4.63	88.8	3.8	39.4
POLAR	zipfian	Off	On	Off	3.171	2.70	1.92	3.11	3.13	52.5	63.8	36.2
POLAR	zipfian	Off	On	On	3.180	2.75	1.95	3.12	3.15	46.3	40.0	35.2
POLAR	zipfian	On	Off	Off	3.340	2.83	3.77	3.28	5.64	98.8	1.3	33.8
POLAR	zipfian	On	Off	On	3.352	2.85	3.13	3.36	5.23	90.0	0.0	33.5
POLAR	zipfian	On	On	Off	3.177	2.72	1.94	3.12	3.14	91.3	67.5	31.8
POLAR	zipfian	On	On	On	3.192	2.74	2.00	3.13	3.15	40.0	26.3	31.8
NOPE	baseline	Off	Off	Off	3.224	2.76	3.71	3.16	6.68	97.5	1.3	41.5
NOPE	baseline	Off	Off	On	3.219	2.75	2.76	3.17	5.49	92.5	10.0	36.5
NOPE	baseline	Off	On	Off	3.140	2.66	2.34	3.09	3.23	97.5	16.3	36.8
NOPE	baseline	Off	On	On	3.150	2.70	2.14	3.09	3.24	81.3	0.0	34.2
NOPE	baseline	On	Off	Off	3.209	2.74	2.99	3.15	6.13	90.0	18.8	32.5
NOPE	baseline	On	Off	On	3.212	2.77	2.90	3.17	5.65	81.3	0.0	30.3
NOPE	baseline	On	On	Off	3.149	2.67	2.35	3.10	3.28	80.0	16.3	30.0
NOPE	baseline	On	On	On	3.147	2.68	2.09	3.09	3.23	88.8	21.3	28.2
NOPE	weak	Off	Off	Off	3.214	2.77	3.10	3.16	6.34	95.0	0.0	40.9

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Table 8: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	weak	Off	Off	On	3.222	2.75	3.05	3.19	5.81	81.3	0.0	36.1
NOPE	weak	Off	On	Off	3.147	2.68	2.07	3.09	3.19	88.8	21.3	37.5
NOPE	weak	Off	On	On	3.142	2.68	2.12	3.08	3.23	82.5	0.0	33.7
NOPE	weak	On	Off	Off	3.215	2.75	4.87	3.16	12.48	95.0	5.0	32.0
NOPE	weak	On	Off	On	3.221	2.73	2.63	3.18	4.84	81.3	0.0	29.3
NOPE	weak	On	On	Off	3.147	2.67	2.48	3.09	3.23	97.5	1.3	30.8
NOPE	weak	On	On	On	3.148	2.68	2.14	3.09	3.24	93.8	1.3	28.4
NOPE	strong	Off	Off	Off	3.231	2.74	4.88	3.18	11.85	90.0	2.5	38.4
NOPE	strong	Off	Off	On	3.214	2.72	2.68	3.17	5.28	82.5	0.0	36.4
NOPE	strong	Off	On	Off	3.144	2.68	2.37	3.09	3.24	96.3	6.3	35.7
NOPE	strong	Off	On	On	3.150	2.70	2.10	3.10	3.25	93.8	7.5	33.8
NOPE	strong	On	Off	Off	3.207	2.72	3.13	3.15	6.20	73.8	0.0	31.4
NOPE	strong	On	Off	On	3.214	2.74	2.49	3.17	4.36	86.3	8.8	28.9
NOPE	strong	On	On	Off	3.146	2.67	2.24	3.09	3.23	83.8	17.5	29.1
NOPE	strong	On	On	On	3.147	2.68	2.16	3.09	3.28	96.3	0.0	27.5
NOPE	discrete	Off	Off	Off	3.202	2.75	3.33	3.15	6.23	85.0	1.3	38.8
NOPE	discrete	Off	Off	On	3.234	2.79	3.11	3.19	5.47	70.0	18.8	37.0
NOPE	discrete	Off	On	Off	3.145	2.68	2.14	3.09	3.27	98.8	16.3	36.0
NOPE	discrete	Off	On	On	3.145	2.68	2.10	3.09	3.25	95.0	3.8	34.3
NOPE	discrete	On	Off	Off	3.212	2.88	3.04	3.16	6.88	42.5	0.0	31.2
NOPE	discrete	On	Off	On	3.211	2.78	2.73	3.16	5.28	82.5	1.3	29.1
NOPE	discrete	On	On	Off	3.145	2.68	2.27	3.08	3.26	91.3	11.3	30.3
NOPE	discrete	On	On	On	3.149	2.68	2.17	3.08	3.32	83.8	0.0	28.6
NOPE	zipfian	Off	Off	Off	3.209	2.71	3.18	3.15	6.40	90.0	0.0	40.8
NOPE	zipfian	Off	Off	On	3.219	2.75	2.71	3.17	5.44	77.5	1.3	36.1
NOPE	zipfian	Off	On	Off	3.141	2.67	2.55	3.08	3.38	96.3	0.0	38.0
NOPE	zipfian	Off	On	On	3.150	2.69	2.11	3.09	3.25	90.0	3.8	33.7
NOPE	zipfian	On	Off	Off	3.207	2.77	3.32	3.15	7.09	82.5	0.0	31.3
NOPE	zipfian	On	Off	On	3.227	2.79	2.91	3.18	5.94	93.8	11.3	30.2
NOPE	zipfian	On	On	Off	3.146	2.68	2.39	3.10	3.25	96.3	21.3	29.6

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Table 8: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	zipfian	On	On	On	3.148	2.68	2.17	3.09	3.28	90.0	6.3	28.7
ROPE	baseline	Off	Off	Off	3.159	2.85	3.36	3.09	5.21	42.5	0.0	40.4
ROPE	baseline	Off	Off	On	3.163	2.86	3.79	3.17	5.86	15.0	0.0	37.2
ROPE	baseline	Off	On	Off	3.145	2.81	2.86	3.09	3.57	73.8	0.0	37.1
ROPE	baseline	Off	On	On	3.151	2.80	2.95	3.12	3.72	13.8	0.0	34.4
ROPE	baseline	On	Off	Off	3.163	3.00	3.47	3.10	5.46	38.8	0.0	33.2
ROPE	baseline	On	Off	On	3.167	2.88	4.06	3.16	5.92	0.0	0.0	30.2
ROPE	baseline	On	On	Off	3.153	2.73	2.45	3.09	3.57	75.0	0.0	31.4
ROPE	baseline	On	On	On	3.154	2.83	2.86	3.12	3.72	32.5	0.0	29.4
ROPE	weak	Off	Off	Off	3.158	2.88	3.46	3.10	5.36	46.3	0.0	41.4
ROPE	weak	Off	Off	On	3.163	2.84	3.96	3.18	5.93	12.5	0.0	37.8
ROPE	weak	Off	On	Off	3.149	2.71	2.46	3.09	3.56	37.5	0.0	36.5
ROPE	weak	Off	On	On	3.156	2.85	2.77	3.12	3.60	23.8	0.0	34.9
ROPE	weak	On	Off	Off	3.155	2.83	3.49	3.10	5.27	38.8	0.0	32.3
ROPE	weak	On	Off	On	3.157	3.05	4.12	3.18	5.86	8.8	0.0	30.5
ROPE	weak	On	On	Off	3.149	2.77	2.76	3.09	3.56	57.5	0.0	31.1
ROPE	weak	On	On	On	3.149	2.77	3.13	3.12	3.80	17.5	0.0	28.0
ROPE	strong	Off	Off	Off	3.165	2.78	3.34	3.10	5.27	30.0	0.0	39.0
ROPE	strong	Off	Off	On	3.169	2.90	3.68	3.19	5.80	31.3	0.0	36.3
ROPE	strong	Off	On	Off	3.158	2.85	2.63	3.10	3.50	51.3	0.0	35.9
ROPE	strong	Off	On	On	3.162	2.86	3.08	3.14	3.76	10.0	0.0	33.2
ROPE	strong	On	Off	Off	3.161	2.85	3.58	3.10	5.28	47.5	0.0	31.1
ROPE	strong	On	Off	On	3.165	2.85	4.03	3.19	5.90	2.5	0.0	29.2
ROPE	strong	On	On	Off	3.155	2.80	2.70	3.10	3.62	61.3	0.0	29.3
ROPE	strong	On	On	On	3.162	2.85	3.08	3.13	3.74	2.5	0.0	28.4
ROPE	discrete	Off	Off	Off	3.165	2.91	3.43	3.11	5.42	22.5	0.0	39.7
ROPE	discrete	Off	Off	On	3.158	2.84	4.05	3.15	5.90	3.8	0.0	36.9
ROPE	discrete	Off	On	Off	3.159	2.83	2.67	3.11	3.56	47.5	0.0	36.3
ROPE	discrete	Off	On	On	3.161	2.79	2.71	3.13	3.74	41.3	0.0	33.7
ROPE	discrete	On	Off	Off	3.171	2.79	3.45	3.12	5.36	51.3	0.0	31.5

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Table 8: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
ROPE	discrete	On	Off	On	3.164	2.80	3.74	3.16	5.86	21.3	0.0	30.0
ROPE	discrete	On	On	Off	3.162	2.74	2.47	3.11	3.53	50.0	0.0	30.9
ROPE	discrete	On	On	On	3.154	2.79	2.68	3.11	3.67	18.8	0.0	28.9
ROPE	zipfian	Off	Off	Off	3.163	2.82	3.25	3.10	5.22	62.5	0.0	39.8
ROPE	zipfian	Off	Off	On	3.161	2.94	3.94	3.18	5.81	16.3	0.0	37.1
ROPE	zipfian	Off	On	Off	3.149	2.78	2.70	3.09	3.63	46.3	0.0	36.7
ROPE	zipfian	Off	On	On	3.151	2.80	2.63	3.11	3.68	2.5	0.0	33.9
ROPE	zipfian	On	Off	Off	3.161	2.78	3.38	3.10	5.32	47.5	0.0	32.6
ROPE	zipfian	On	Off	On	3.160	3.01	4.11	3.17	5.81	17.5	0.0	30.2
ROPE	zipfian	On	On	Off	3.149	2.73	2.39	3.09	3.55	62.5	0.0	29.8
ROPE	zipfian	On	On	On	3.152	2.79	2.99	3.12	3.78	26.3	0.0	28.5

F Retrieval Breakdown

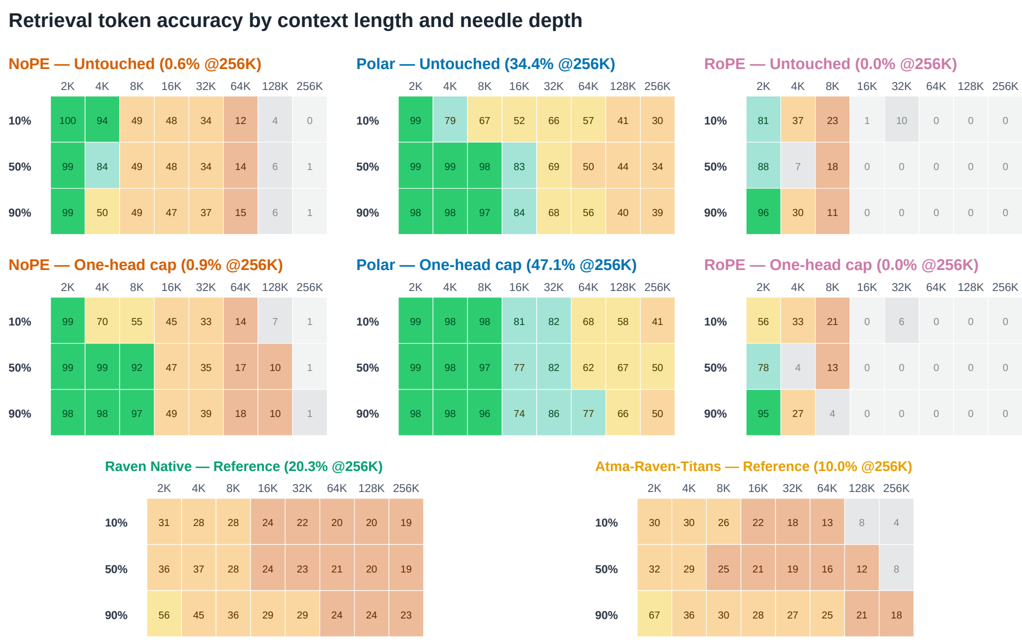

Figure 5: Retrieval depth sweeps for the three matched attention variants before and after the one-head retention cap, followed by untouched Raven Native and Atma-Raven-Titans reference panels. Each cell is teacher-forced token accuracy averaged over passkey and NIAH and over synthetic and FinePDFs haystacks. The Raven rows are separately optimized comparisons, not clamped ablations.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	60.7	51.1	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 9: Zero-shot retrieval accuracy (%) by haystack type and context length, averaged over needle depths.

Figure 5 exposes depth-wise variation. Tables 9 and 10 separate partial target signal from exact five-token success. For Polar, the 64K exact rate is 65.7% on synthetic contexts (95% Wilson interval 60.1–70.8%) and 0/300 on FinePDFs (upper Wilson endpoint 1.3%). At 256K the corresponding exact rates are 18.0% (14.1–22.7%) and 0/300. These results support exact synthetic extrapolation and teacher-forced real-text signal, but not exact real-text retrieval.

Model	Haystack	Token accuracy (%)			Exact five-token accuracy (%)		
		2K	64K	256K	2K	64K	256K
Polar	Synthetic	99.6	92.3	68.4	96.3	65.7	18.0
	FinePDFs	98.6	16.3	0.3	93.0	0.0	0.0
NoPE	Synthetic	99.5	26.5	1.2	95.3	4.7	0.0
	FinePDFs	99.3	0.0	0.0	96.7	0.0	0.0
RoPE	Synthetic	81.7	0.1	0.0	42.3	0.0	0.0
	FinePDFs	94.8	0.0	0.0	77.3	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	35.8	19.3	8.0	0.3	0.0
	FinePDFs	23.2	0.5	0.7	4.7	0.0	0.0
Raven Native	Synthetic	55.9	40.9	35.1	2.0	0.0	0.0
	FinePDFs	22.1	2.9	5.4	0.0	0.0	0.0

Table 10: Teacher-forced retrieval under the two archived metrics. Token accuracy can expose partial target signal, whereas exact accuracy requires the entire five-token value to be predicted correctly under teacher forcing. Each haystack–length entry averages 300 sequences (two tasks, three depths, 50 trials).

F.1 Open-weight context

Table 11 reports the ten archived open-weight controls. They differ in training data, tokenizer, architecture, parameter count, and native context length; several were trained at lengths beyond 2K. They are therefore practical context, not matched length-extrapolation baselines and not evidence of a Polar advantage over the corresponding model families.

Open-weight model	NLL 2K	NLL 64K	Needle 2K	Needle 64K
SmolLM2-360M	1.99	6.04	100.0	8.1
LFM2.5-230M	3.01	6.69	38.1	19.4
LFM2.5-350M	3.18	6.63	45.6	46.3
Qwen2.5-0.5B	2.03	1.53	100.0	38.1
Qwen3-0.6B-Base	1.95	1.40	100.0	95.0
Qwen3.5-0.8B-Base	1.94	1.33	100.0	100.0
Gemma-3-270M	2.48	1.90	96.2	24.4
Granite-4.0-350M	2.52	3.02	40.6	8.8
Granite-4.0-H-350M	2.22	2.06	75.0	54.4
Falcon-H1-0.5B	1.99	2.54	92.5	19.4

Table 11: Open-weight controls evaluated through 64K. NLL is fixed-target FinePDFs nats/token; Needle is teacher-forced target-token accuracy over 16 trials. These checkpoints are not matched for data, tokenizer, native context, or training budget.

G Short-Context Quality and Systems

Group	Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
Matched	NoPE	30.16	37.88	66.27	51.93	50.53	32.44	31.60	60.37	45.15
Matched	RoPE	30.25	37.19	66.76	50.67	49.12	29.43	32.40	60.95	44.60
Matched	Polar	28.47	36.19	66.87	51.62	49.12	26.09	31.80	56.21	43.29
Raven/AdamW	Atma-Raven-Titans	26.00	34.12	65.56	52.49	48.07	27.76	29.00	61.41	43.05
Raven/AdamW	Raven Native	27.48	33.56	64.69	51.70	49.12	27.09	28.80	61.93	43.05

Table 12: Task-level zero-shot accuracy (%) on the eight short-context controls (22,939 examples per model). LAMBADA, WinoGrande, and BoolQ use standard accuracy; HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and OpenBookQA use length-normalized accuracy. Mean is the unweighted average of the eight displayed metrics; bold is best within each comparison group.

Table 12 gives the exact per-task controls shown in main-text Figure 3. Figure 6 shows the serving-length trend, and Table 13 summarizes the quality–systems trade-off.

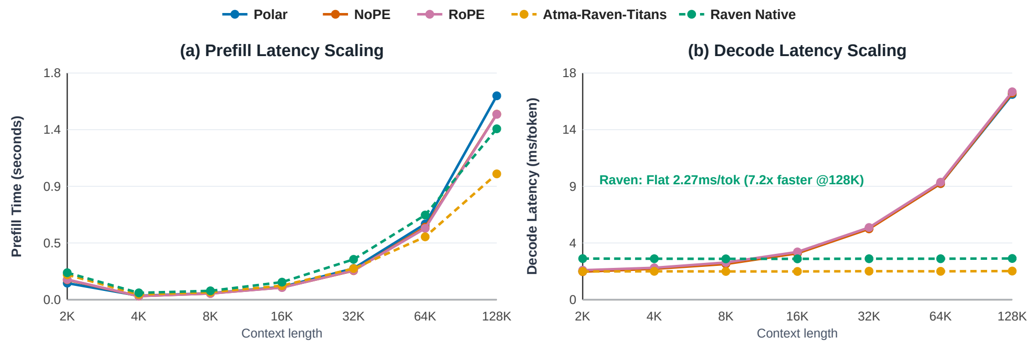

Figure 6: Single-sequence serving on one L40S. Attention decode scales with KV-cache length; recurrent Raven uses fixed state.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 13: Short-context quality and serving trade-offs. Serving uses one sequence and one timing sample; it is descriptive rather than an uncertainty estimate.

H Systems Implementation and Optimization Breakdown

The inference stack uses separate kernels for full-context prefill, paged autoregressive decode, and recurrent-state updates. Table 14 distinguishes what each optimization changes. In particular, streaming removes quadratic score storage but not quadratic prefill arithmetic, while the recurrent Raven path changes the state representation itself and therefore has different scaling.

Path	Optimization	Operational effect
Polar prefill	Block-streamed K, V tiles with online (M, L, Q_2, S) accumulation	Avoids materializing $T \times T$ scores; retains full-context $O(T^2)$ arithmetic.
Polar decode	Paged KV-cache kernel grouping the four query heads that share each KV head	Reuses KV loads across the GQA group; decode remains $O(T)$ in cache length.
Gated-delta step	Fused in-place decay, prediction, delta write, and readout	Avoids materializing per-step intermediate states and preserves $O(1)$ recurrent state size.
Ragged prefill	Packed prompts with grouped compatible tile shapes	Reduces padding work on heterogeneous prompt batches.
Compiler boundary	Recurrent scan and custom kernels exposed as opaque operations	Prevents compiler graph breaks around the stateful update while retaining explicit numerical tests.

Table 14: Implementation-level optimization breakdown. Complexity statements describe the sequence-mixing path and exclude shared MLP and embedding costs.

H.1 Single-sequence scaling

Table 15 gives the underlying single-sequence measurements. On one NVIDIA L40S, attention decode increases from approximately 2.3 ms/token at 2K to 16.3–16.5 ms/token at 128K because every step reads a longer KV cache. The recurrent models remain nearly flat because they update fixed-size state. Prefill is less monotonic at short lengths because launch overhead and kernel occupancy dominate before the matrix operations reach their efficient regime. These timings use batch size 1, 32 generated tokens, and one recorded timing sample per cell. They are descriptive measurements, not confidence intervals.

Table 15: Detailed prefill and decode scaling from 2K to 128K tokens on one NVIDIA L40S GPU (batch size 1, 32 generated tokens).

Model	Metric	2K	4K	8K	16K	32K	64K	128K
Polar	Prefill latency (s)	0.131	0.031	0.051	0.101	0.248	0.601	1.619
	Prefill throughput (tok/s)	15,640	133,884	160,145	161,551	132,091	109,050	80,959
	Decode latency (ms/tok)	2.27	2.49	2.88	3.70	5.66	9.22	16.30
	Decode throughput (tok/s)	441.2	402.3	346.6	270.1	176.8	108.5	61.3
NoPE	Prefill latency (s)	0.157	0.028	0.049	0.097	0.230	0.576	1.473
	Prefill throughput (tok/s)	13,070	146,568	166,504	168,510	142,319	113,810	89,013
	Decode latency (ms/tok)	2.22	2.45	2.83	3.68	5.62	9.22	16.40
	Decode throughput (tok/s)	450.1	408.8	353.0	271.7	177.8	108.4	61.0
RoPE	Prefill latency (s)	0.160	0.028	0.049	0.096	0.232	0.566	1.472
	Prefill throughput (tok/s)	12,761	147,089	168,287	170,936	141,249	115,786	89,015
	Decode latency (ms/tok)	2.34	2.53	2.95	3.79	5.73	9.33	16.51
	Decode throughput (tok/s)	427.4	395.3	338.7	263.7	174.5	107.2	60.6
Atma-Raven-Titans	Prefill latency (s)	0.198	0.041	0.059	0.111	0.247	0.499	0.999
	Prefill throughput (tok/s)	10,334	100,989	137,903	147,434	132,916	131,464	131,195
	Decode latency (ms/tok)	2.24	2.25	2.24	2.24	2.26	2.25	2.27
	Decode throughput (tok/s)	446.6	445.1	446.0	447.1	442.6	444.8	440.3
Raven Native	Prefill latency (s)	0.214	0.054	0.070	0.139	0.320	0.672	1.357
	Prefill throughput (tok/s)	9,558	75,738	116,378	117,605	102,354	97,527	96,559
	Decode latency (ms/tok)	3.26	3.25	3.24	3.24	3.25	3.25	3.27
	Decode throughput (tok/s)	306.4	307.5	308.2	308.3	307.6	307.5	306.1

H.2 Production-framework stress comparison

To test whether the custom kernels introduce an obvious systems bottleneck at a larger shape, we configured ATMA's inference engine with a 9.2B-parameter stress shape and compared it with vLLM 0.25.0 serving Qwen3.5-9B on the same L40S in BF16 (Kwon et al., 2023). The models are not weight- or architecture-matched, so Table 16 is a backend stress comparison rather than a model-quality or universal serving claim. ATMA trails on short homogeneous prefill, is similar on long heterogeneous prefill, and leads the recorded high-batch decode cells.

Phase	Batch/workload	ATMA engine	vLLM/Qwen3.5-9B	Ratio
Prefill	Homogeneous $B = 8, T = 512$ (4,096 tok)	11,947 tok/s	12,772 tok/s	0.94×
	Heterogeneous short-heavy (816 tok)	5,752 tok/s	8,002 tok/s	0.72×
	Heterogeneous mixed (4,384 tok)	12,779 tok/s	11,996 tok/s	1.07×
	Heterogeneous long-tail (8,192 tok)	12,251 tok/s	11,721 tok/s	1.05×
Decode, context 512	Batch 128	2,939 tok/s	2,247 tok/s	1.31×
	Batch 256	4,501 tok/s	2,921 tok/s	1.54×
	Batch 512	5,919 tok/s	2,558 tok/s	2.31×
	Heterogeneous $B = 128$ (context 64–1536)	2,950 tok/s	2,403 tok/s	1.23×

Table 16: Recorded systems-scale throughput for ATMA's 9.2B stress shape and vLLM 0.25.0 serving Qwen3.5-9B on one L40S (BF16, greedy sampling). Ratio is ATMA/vLLM.

I Retention-Horizon Diagnostic

This section records the complete post-hoc chain from parameter inspection to re-evaluation. It is an analysis of released checkpoints, not an additional trained architecture. The primary checkpoint results in Section 5 remain untouched; every cap below is opt-in, applied only at inference, and removed after each condition.

I.1 Parameter-only operating points

The memory gate has one learned row per query/memory head, rather than one scalar per attention KV head. For each row we decompose the logit as

$$z_{\gamma,t} = W_\gamma x_t + b_\gamma, \qquad \gamma_0 = \sigma(b_\gamma), \qquad H_{1/2}(\gamma_0) = \frac{\log(1/2)}{\log \gamma_0}. \tag{11}$$

Here $x = 0$ defines a reproducible learned operating point. It is not the empirical mean runtime gate: every reported outlier has a nonzero W_γ row, so activations can shorten or lengthen its instantaneous horizon. Table 17 reports the maximum over all four memory blocks and eight heads per block.

Checkpoint	Block/head	Maximum $H_{1/2}$	Role
NoPE, L4, microbatch 4	2/7	185.3	Earlier control
NoPE, L40S, microbatch 4	2/2	1.38M	Microbatch control
NoPE, L40S, microbatch 16	2/5	21.01M	Promoted matched model
Polar, L40S, microbatch 16	2/6	3.07M	Promoted matched model
RoPE, L40S, microbatch 16	2/1	682.2	Matched negative control
Atma-Raven-Titans	2/3	41.4	Recurrent-family reference

Table 17: Largest learned zero-input retention half-life in each base checkpoint. "Block/-head" uses zero-based indices. The full scan reports every layer/head, logit, γ_0, $1/e$ horizon, and W_γ row norm.

Scanning the separately fine-tuned BABILong checkpoints gives 20.86M tokens for NoPE, 3.05M for Polar, and 680 for RoPE. Thus the selected outlier survives adaptation nearly unchanged; each BABILong cap is nevertheless selected afresh from the final fine-tuned checkpoint rather than copied from the base model.

I.2 Runtime intervention

For a target maximum half-life $H_{\max}$, the diagnostic clips the final activation-conditioned gate logit,

$$z'_{\gamma,t} = \min\Big(z_{\gamma,t}, \mathrm{logit}(2^{-1/H_{\max}})\Big), \qquad \gamma'_t = \sigma(z'_{\gamma,t}). \tag{12}$$

All experiments use $H_{\max} = 256$, hence $\gamma'_t \leq 0.997296$. Only the single layer/head with the largest parameter-only b_γ is targeted. This upper bound leaves smaller token-conditioned gates unchanged, modifies neither checkpoint tensors nor attention, and is removed after the condition. It therefore tests whether excessive retention mediates a failure; it does not test a train-time parameterization.

The first sweep is paired within one process: baseline and cap share the loaded model, nested FinePDFs documents, generated needle examples, seeds, and SDPA implementation. Four documents and eight needle trials are evaluated at each length from 2K to 256K. Table 18 reports the longest endpoint.

Checkpoint	Clean NLL at 256K	Needle CE at 256K	Needle acc. at 256K
NoPE, L4 mbs4	3.615 → 2.776	3.730 → 4.360	15.0 → 10.0
NoPE, L40S mbs4	9.589 → 1.181	10.446 → 6.108	0.0 → 0.0
NoPE, L40S mbs16	12.003 → 1.138	12.198 → 6.051	0.0 → 0.0
Polar, L40S mbs16	1.252 → 1.057	2.387 → 1.707	50.0 → 65.0
RoPE, L40S mbs16	2.948 → 2.754	6.157 → 6.019	0.0 → 0.0

Table 18: Same-process paired pilot, baseline → one-head 256-token half-life cap. Accuracy is teacher-forced target-token accuracy (%). The L4 and RoPE rows show that likelihood improvements need not restore retrieval.

At 2K, clean NLL changes by +0.001, +0.013, +0.022, +0.010, and +0.004 for the rows in table order. The cap therefore leaves ordinary likelihood nearly unchanged in this pilot. It is not cost-free: RoPE's 2K needle accuracy falls from 82.5% to 50.0%, and the L4 NoPE 256K needle result also worsens. These controls argue against treating 256 tokens as a universal deployment setting.

I.3 Independent benchmark re-evaluation

After the pilot, we evaluate the capped promoted checkpoints on all eight short-context tasks, the full synthetic/FinePDFs retrieval matrix, three fixed-target long-document datasets, and BABILong QA1–QA10. Base-model and BABILong-fine-tuned checkpoints are pinned to their archived Hugging Face revisions. The BABILong checkpoints are not fine-tuned again; the cap is installed only for held-out evaluation. Direct checkpoint-exact scoring is used throughout because the paged serving path does not expose the diagnostic hook. All 15 model–suite jobs complete without OOM.

Tables 19–23 report every evaluated length and every short-context task, rather than only the 256K endpoint. The broad run reuses pinned archived baselines; Table 18 is the controlled same-process comparison.

Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
NoPE	30.16 → 30.22	37.88 → 37.88	66.27 → 66.21	51.93 → 51.38	50.53 → 50.70	32.44 → 32.44	31.60 → 31.60	60.37 → 60.34	45.15 → 45.10
Polar	28.47 → 28.22	36.19 → 36.20	66.87 → 67.30	51.62 → 51.38	49.12 → 48.60	26.09 → 26.42	31.80 → 32.00	56.21 → 55.87	43.29 → 43.25
RoPE	30.25 → 30.16	37.19 → 37.17	66.76 → 66.97	50.67 → 51.07	49.12 → 48.95	29.43 → 29.77	32.40 → 32.20	60.95 → 61.01	44.60 → 44.66

Table 19: Complete short-context control, untouched → capped (accuracy, %). The intervention changes no task mean materially.

NoPE's BPB recovery begins immediately beyond the training length and remains nearly flat through 256K: mean BPB is 1.454 at 4K and 1.595 at 256K after clamping, versus 1.717 and 8.097 untouched. At 256K its dataset BPB changes from 11.666 to 1.180 on FinePDFs, 4.902 to 1.224 on PG-19, and 7.723 to 2.382 on Proof-Pile. Its mean degradation relative

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	99.2 → 98.8	75.8 → 88.7	48.9 → 81.1	47.5 → 46.7	35.2 → 35.5	13.2 → 16.6	5.1 → 9.2	0.6 → 0.9
Polar	98.9 → 98.6	92.4 → 97.8	87.5 → 97.2	73.1 → 77.6	67.8 → 83.5	54.3 → 69.0	41.8 → 63.6	34.4 → 47.1
RoPE	88.2 → 76.1	24.8 → 21.5	17.4 → 13.0	0.4 → 0.0	3.4 → 2.1	0.1 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 20: Complete retrieval curve, untouched → capped, reported as teacher-forced target-token accuracy (%). Each entry averages passkey and NIAH, synthetic and FinePDFs haystacks, and three depths.

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	96.0 → 94.0	55.8 → 75.0	44.3 → 62.2	39.0 → 32.7	11.8 → 9.2	2.3 → 0.5	0.3 → 0.0	0.0 → 0.0
Polar	94.7 → 93.2	79.8 → 89.7	74.5 → 86.5	49.0 → 43.0	38.0 → 44.2	32.8 → 33.0	13.2 → 19.8	9.0 → 16.3
RoPE	59.8 → 45.0	2.2 → 0.8	0.3 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 21: Complete retrieval curve, untouched → capped, reported as exact five-token accuracy (%). Each entry averages passkey and NIAH, synthetic and FinePDFs haystacks, and three depths.

Model	0K	1K	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	63 → 63	57 → 60	55 → 55	56 → 56	39 → 57	21 → 54	6 → 45	0 → 37	0 → 35	0 → 39
Polar	55 → 55	54 → 54	52 → 52	54 → 54	56 → 57	50 → 50	46 → 43	45 → 44	35 → 44	28 → 42
RoPE	59 → 61	67 → 68	64 → 66	43 → 50	28 → 33	19 → 21	15 → 17	16 → 15	14 → 13	14 → 11

Table 22: Complete BABILong curve after the same 0K/1K/2K adaptation, untouched → capped (macro exact match, %). Checkpoints are not fine-tuned again; the cap is installed only for held-out evaluation.

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	1.432 → 1.454	1.717 → 1.454	2.656 → 1.473	3.374 → 1.484	5.527 → 1.504	5.885 → 1.506	6.369 → 1.530	8.097 → 1.595
Polar	1.451 → 1.458	1.442 → 1.451	1.406 → 1.407	1.417 → 1.384	1.500 → 1.399	1.569 → 1.400	1.690 → 1.449	1.825 → 1.501
RoPE	1.439 → 1.440	1.642 → 1.656	1.783 → 1.783	1.935 → 1.932	2.094 → 2.082	2.252 → 2.220	2.413 → 2.378	2.512 → 2.460

Table 23: Complete fixed-target likelihood curve, untouched → capped (mean bits per byte; lower is better), averaged over FinePDFs, PG-19, and Proof-Pile.

to 2K falls from 5.65× to 1.10×. BABILong shows a parallel length-dependent recovery: untouched accuracy falls from 21% at 16K to 6/0/0/0% at 32/64/128/256K, whereas the cap yields 54/45/37/35/39%. Exact retrieval nevertheless remains zero at 256K. Thus the intervention restores usable recurrent computation and likelihood without restoring every form of direct lookup.

For Polar, token/exact retrieval at 64K changes from 54.3/32.83% to 69.0/33.0%, at 128K from 41.77/13.17% to 63.57/19.83%, and at 256K from 34.37/9.00% to 47.07/16.33%. The 256K synthetic component changes from 68.4/18.0% to 77.47/32.67%; FinePDFs token accuracy changes from 0.33% to 16.67%, while exact FinePDFs retrieval remains zero. Mean BPB degradation falls from 1.26× to 1.03×.

RoPE is the decisive boundary: its maximum parameter-only half-life is three to four orders of magnitude below the NoPE and Polar outliers, and the same cap neither restores retrieval nor improves BABILong. Across all three models, downstream means move by at most 0.06 points. The evidence therefore supports a selective claim: an isolated near-unit retention gate is a major causal mediator of the observed NoPE and Polar extrapolation degradation. It does not explain every failure mode, identify the training event that produced the outlier, or validate a particular bounded-gate training formulation.

The complete machine-readable artifacts are under gamma_diagnostics/results/: parameters/gamma_parameters.json for the scan, all_clamp_sweep_262k.json for the paired pilot, and re_evaluation/run-summary.json plus per-suite logs for the full benchmark. The executable protocol and reporting caveats are in gamma_diagnostics/README.md.

J Stress and Environment Diagnostics

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84
B6	1.02	1.02	1.03	1.05	1.07	1.09	1.10	1.11
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17

RoPE: Representational Attenuation

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94
B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92
B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98
B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03
B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92
B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00
B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99
B8	1.01	1.00	1.00	1.00	0.99	0.99	0.99	0.99
B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96
B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04
B11	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00
B12	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00
B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00
B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25
B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16

NoPE (L40S mbs16): Block 7 Yield Locus

	2K	4K	8K	16K	32K	64K	128K	256K
B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.16	1.14	1.12	1.12	1.12	1.11	1.11	1.11
B11	1.04	1.03	1.03	1.02	1.01	1.01	1.00	1.00
B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 7: Block-wise sampled secant gains for the three matched attention checkpoints over 64 nested FinePDFs documents and 64 perturbation directions per document. Raven Native and Atma-Raven-Titans were not run under this checkpoint-specific probe. The diagnostic localizes changes but does not estimate a global Lipschitz constant.

Figure 7 reports the sampled gains. For each block f_i, the diagnostic samples

$$G_i = \frac{\|f_i(x + \delta x) - f_i(x)\|_{\mathrm{rms}}}{\|\delta x\|_{\mathrm{rms}}}, \qquad \|\delta x\|_{\mathrm{rms}} = 0.02\|x\|_{\mathrm{rms}}. \tag{13}$$

The maximum sampled gain is a lower bound on the local operator norm along sampled directions. It is not a tight bound on the block Lipschitz constant. In the observed NoPE collapse, changes first concentrate near blocks 6–7 and later propagate through the residual stream. Polar's sampled gains and activation moments remain comparatively flat through 256K.

K Checkpoint-Variability Diagnostic Journey

The variability finding emerged sequentially. Stage I used L4 GPUs, followed by the first 10B NoPE control on an L4. The anomaly became visible only after the remaining training moved to L40S: short-context convergence remained nearly unchanged while the 256K extrapolation results diverged. Table 24 records the subsequent elimination sequence. The recorded run configurations do not provide a paired controlled seed, so random initialization and the resulting stochastic trajectory remain confounded with the infrastructure transition.

All five headline models were trained in the same L40S software and hardware environment, so their primary comparison is internally matched. The earlier L4 checkpoint is retained only as a post-hoc case study. It motivates a broader audit principle: long-context extrapolation claims should report device class, kernels, seeds, and extended-context diagnostics rather than infer robustness from short-context loss alone. The present evidence does not identify CUDA reduction order, floating-point accumulation, device class, or any other single mechanism as causal because the L4/L40S runs were neither initialized from a recorded common state nor replicated across seeds and kernel configurations.

Step	Observation or intervention	Diagnostic implication
L4 baseline	Stage I and the first 10B NoPE control used L4, microbatch 4; 256K NLL was 4.07.	Established the original extrapolation behavior, but not a replicated device-class estimate.
L40S transition	The promoted NoPE run used L40S, microbatch 16; 2K validation ended at 2.8126 versus 2.8160 on L4, but 256K NLL rose to 10.77.	Revealed that nearly coincident short-context curves did not predict the long-context endpoint.
Evaluation replay	L4 and L40S checkpoints were re-evaluated under PyTorch 2.12 and 2.13 with unchanged scores.	Excluded evaluation-version mismatch for these checkpoints.
Microbatch control	NoPE was retrained on L40S with microbatch 4; 2K validation ended at 2.8128 and 256K NLL at 9.90.	Excluded microbatch size alone as the explanation in this comparison.
Deterministic fixtures	Synthetic 2K stream fixtures across the tested versions and microbatches showed no gradient or loss divergence.	Found no short-context execution discrepancy; did not test the full stochastic training trajectory.

Table 24: Diagnostic journey from the L4 control to the L40S follow-ups. Each implication is limited to the intervention actually performed.